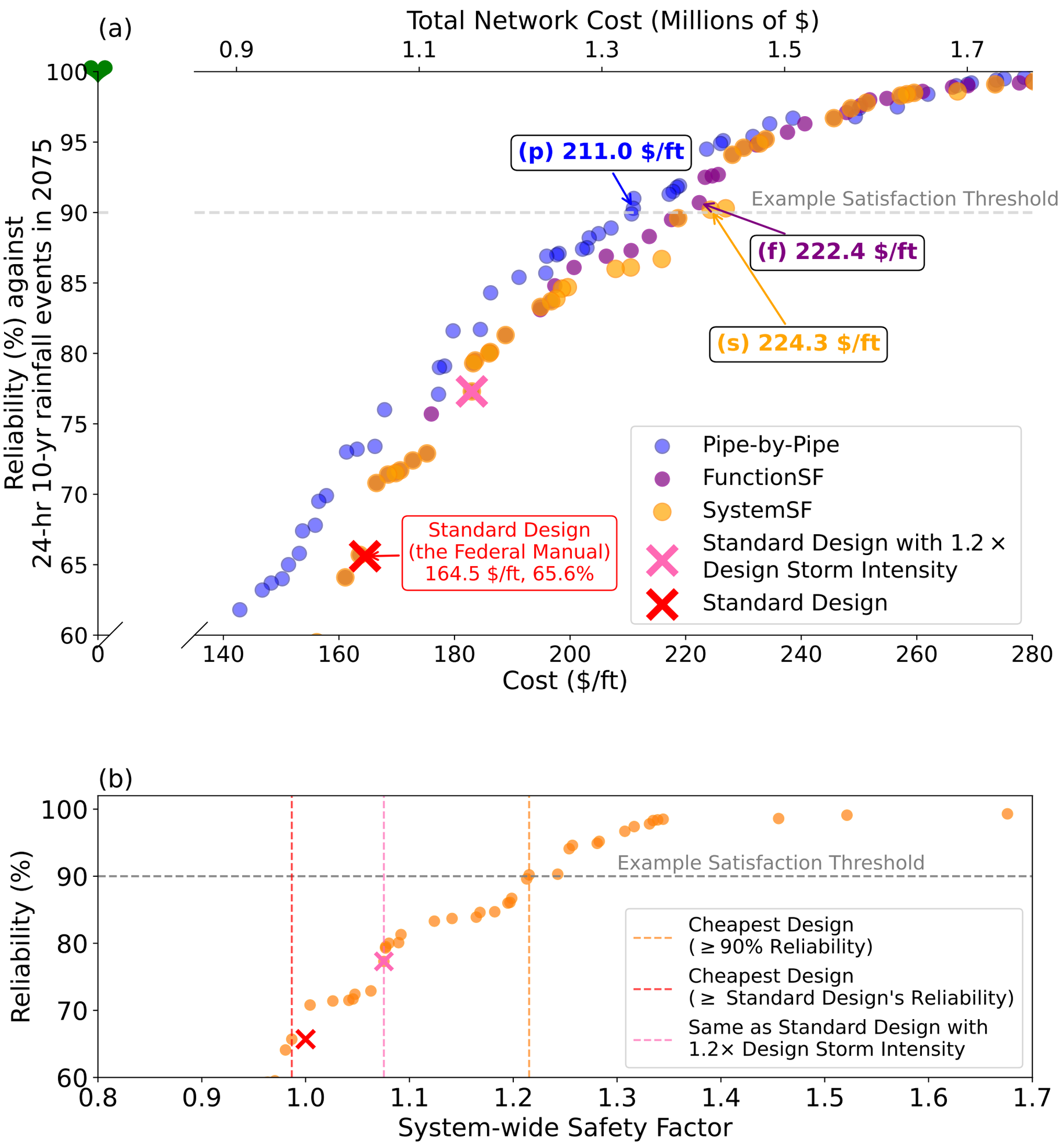
(a)
Total Network Cost (Millions of $)
0.9
1.1
1.3
1.5
1.7
100
95
90
85
80
75
70
65
60
Reliability (%) against
24-hr 10-yr rainfall events in 2075
(p) 211.0 $/ft
Example Satisfaction Threshold
(f) 222.4 $/ft
(s) 224.3 $/ft
Pipe-by-Pipe
FunctionSF
SystemSF
Standard Design with 1.2 ×
Design Storm Intensity
Standard Design
Standard Design
(the Federal Manual)
164.5 $/ft, 65.6%
0
140
160
180
200
220
240
260
280
Cost ($/ft)
(b)
100
90
80
70
60
Reliability (%)
Example Satisfaction Threshold
Cheapest Design
(≥ 90% Reliability)
Cheapest Design
(≥ Standard Design's Reliability)
Same as Standard Design with
1.2× Design Storm Intensity
0.8
0.9
1.0
1.1
1.2
1.3
1.4
1.5
1.6
1.7
System-wide Safety Factor